PersonaPath

Personalize Dining Recommendation Engine for Yelp

Team 5

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Why PersonaPath?

- Traditional platforms tell you WHAT's nearby, but can't match restaurants to WHO you are as a diner
- PersonaPath bridges that gap, by using review data to deliver recommendations tailored to each user.

Feature	Google Maps	Yelp	✦ PersonaPath
Restaurant Search & Discovery	✓	✓	✓
Star Ratings & Reviews	✓	✓	✓
Category Filter (Cuisine/Price)	✓	✓	✓
Detailed Restaurant Info	✗	Partial	✓
Natural Language Search	✗	✗	✓
Occasion-based Recommendation	✗	✗	✓
Behavioral User Profiling	✗	✗	✓

Project Pipeline & Methodology

Data Processing

- Load ~5GB Yelp JSON files
- Filter to Philadelphia as pilot city
- Prepare 300K+ reviews for modeling

Topic Modeling

- LDA on all reviews to uncover 25 topics
- Label with human themes (e.g. Quick & Fresh Lunch)

Feature Engineering

- Build business & user profiles
- Topic vectors
- Cosine similarity scores

Recommender System

- Rank using JSD scores, EAS quality and sentiment score
- generate explanations using LLM

User Interface

- Create Streamlit chatbot for user prompts
- Return top 10 results with evidence

01 02 03 04 05

What We Built?

Key Metrics

- ~ 1900 Philadelphia restaurants indexed
- 133K user profiles built from review history
- 25 LDA topics discovered from 300K reviews
- Scoring include topic similarity, query intent, and EAS quality
- Top 10 results with LLM generated explanations

Streamlit Chat Interface

PersonaPath

"I want somewhere cozy for a romantic dinner with good wine"

1.Zahav
Israeli · Intimate atmosphere · EAS 9.1
Matches your preference for quiet, distinctive dining.

2.Vernick Food & Drink
American · Cozy · Great wine list · EAS 8.7
Reviewers consistently praise the ambiance and sommelier.

3.Parc Brasserie
French · Romantic · Excellent service · EAS 8.4
A top match for your evening dining persona.

What is the Business Value?

For Yelp:

- Personalized results keep users on the platform longer
- Niche restaraunts get matched to the right users
- More users join platform due to better recommendations.

For Users:

- Less time scrolling to find the best restaurant for their preferences.
- More personal recommendations beyond just the type of restaurant and ratings.
- Higher satisfaction with the recommendations.

How the recommender system perform?

MODEL EVALUATION

Performance Comparison

50 users · K=10 · threshold ≥ 4.0 stars · 80/20 temporal split

MODEL	PRECISION@10	RECALL@10	NDCG@10	HITRATE@10
PersonaPath OUR MODEL	0.0180	0.0737	0.0326	0.18
Baseline — Popular non-personalized	0.0060	0.0115	0.0093	0.06

3.5× NDCG vs popular

3× HitRate vs popular

18% users got a match

LOOKING AHEAD

The Road Ahead

01
Scale to Any City
- Add new cities to the data pipeline

02
Add More Information to Results
- Add current wait times, hours to results

03
Group Consensus Mode
- Add the ability to combine multiple user preferences for group dining experiences.

04
BERTopic Upgrade
- Replace LDA with BERTopic for richer, more precise results

05
Travel & Shopping Domains
- Apply to domains outside of restaurants, such as travel, services, and shopping.

PersonaPath makes every recommendation personal

Thank You